

Homework1 CS5800

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1 Exercise 1

1.1

Since $f(n) = \Theta(g(n))$ then there exist constants c_1, c_2, n_0 such that for all $n \geq n_0$:

$$c_1 g(n) \leq f(n) \leq c_2 g(n).$$

So now if we take the log of each side we get:

$$\log(c_1 g(n)) \leq \log f(n) \leq \log(c_2 g(n)).$$

Using the logarithmic property $\log(ab) = \log a + \log b$, this becomes:

$$\log c_1 + \log g(n) \leq \log f(n) \leq \log c_2 + \log g(n).$$

So this means that the difference between $\log f(n)$ and $\log g(n)$ is bounded by constants, which means that yes $\log f(n) = \Theta(\log g(n))$

1.2

I used the same method here. Exponentiating each side by 2 gives:

$$2^{c_1 g(n)} \leq 2^{f(n)} \leq 2^{c_2 g(n)}.$$

Since the exponential function is monotonically increasing, we know that if $f(n) = \Theta(g(n))$ then $2^{f(n)}$ and $2^{g(n)}$ will also grow at the same rate up to some constant factors. Which means that yes it is also true that $2^{f(n)} = \Theta(2^{g(n)})$.

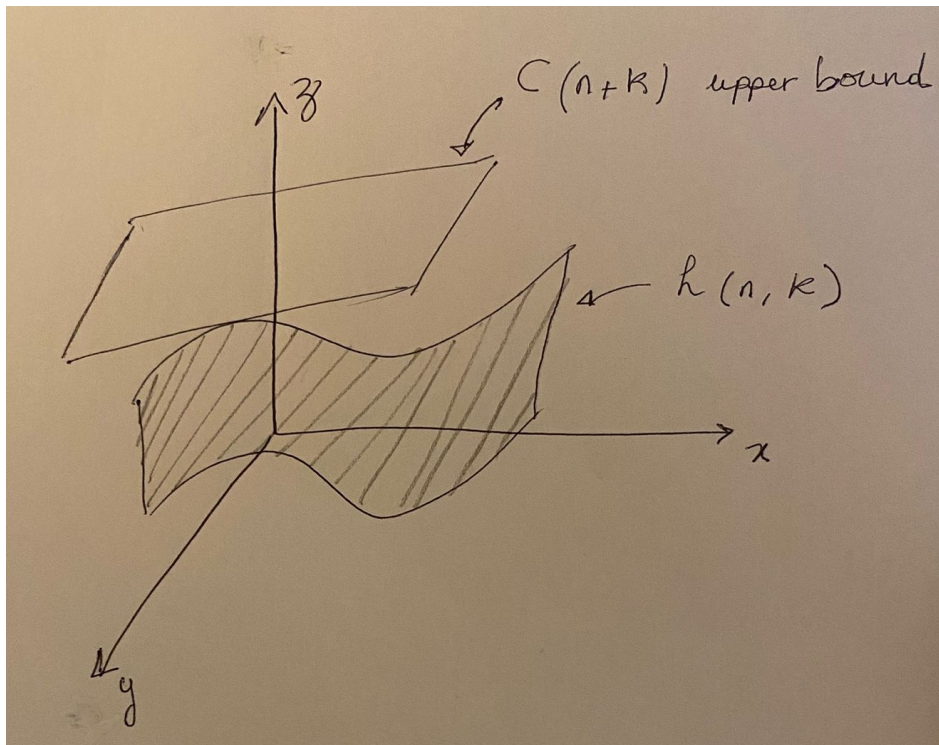
2 Exercise 2

2.1

$h(n, k) = O(n + k)$ means that there exist constants $C > 0, n_0, k_0$ such that for all $n \geq n_0, k \geq k_0$

$$h(n, k) \leq C \cdot (n + k)$$

Mathematically, it means that the function has an upper bound surface (in this case since we are in 3D) instead of line. The following image shows a drawn representation of that:



2.2

We know that $f(n) = O(n)$ and $g(n) = O(n)$. In other words, $f(n) \leq c_1 n$ and $g(k) \leq c_2 k$. And given that c_1 and c_2 are both positive constant, we can simply compare $c_1 n + c_2 k$ to $C(n + k)$. If we get that there exist $C > 0$ such that for all $n \geq n_0, c_1 n + c_2 k \leq C(n + k)$, then automatically $f(n) + g(k) \leq C(n + k)$ since $f(n) + g(k) \leq c_1 n + c_2 k$. The inequality $c_1 n + c_2 k \leq C(n + k)$ ($C > 0$) becomes $c_1 n + c_2 k \leq Cn + Ck$. We have to have $C \geq c_1$ and $C \geq c_2$ for it to hold. So we can simply choose C to be the maximum between both and the inequality would hold: $C = \max(c_1, c_2)$. Therefore there exist $C > 0$ such

that for all $n \geq n_0$: $c_1n + c.c_2k \leq C(n+k)$ ($C > 0$). Which means the for the same conditions $f(n) + cg(k) \leq C(n+k)$. Which finally means that yes it is true that $f(n) + cg(k) = O(n+k)$.

3 Exercise 3

Proof of Big-O

$$f(n) = \sum_{y=1}^n (n^6 y^{23}) = n^6 \sum_{y=1}^n (y^{23}) = n^6 (1^{23} + 2^{23} + 3^{23} + \dots + n^{23})$$

By exaggerating and simplifying, let's first make all the terms in the summation equal to n^{23} (exaggerate). So we would get $f(n) \leq n^6 (n^{23} + n^{23} + n^{23} + \dots + n^{23})$. This gives us: $f(n) \leq n^6 (n \cdot n^{23}) = n^6 (n^{24}) = n^{30}$. So $f(n) \leq Cn^{30}$, we can choose $C=1$. Therefore, $f(n) = O(g(n))$.

Proof of Omega

$$f(n) = n^6 (1^{23} + 2^{23} + 3^{23} + \dots + n^{23})$$

By underestimating all the constants go to 0 and we can change n^{23} to n (since $n^{23} \geq n$):

$$f(n) \geq n^6 (0 + 0 + 0 + \dots + n) = n^6 (n).$$

So now we can say $f(n) \geq n^7$. So $f(n) \geq Cn^7$, we can choose $C=1$. Therefore we also get that $f(n) = \Omega(g(n))$

Finally, we therefore get that $f(n) = O(g(n))$.